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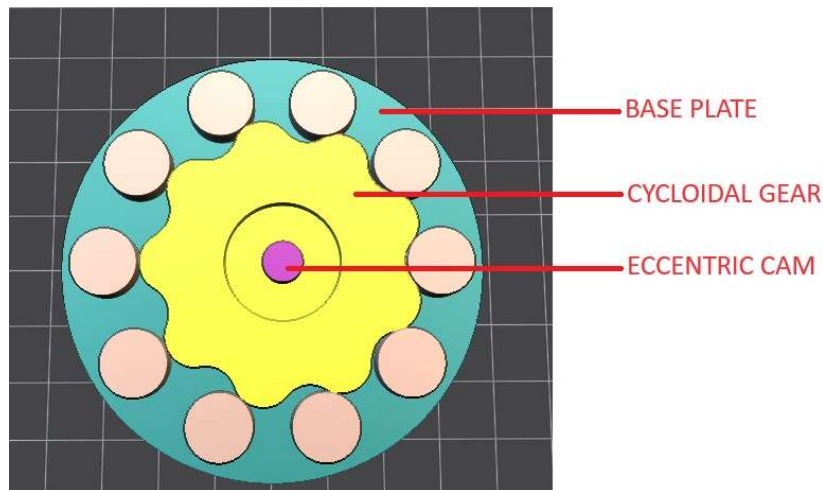
Introduction

The user manual helps to develop the Cycloidal Gear components using Lua scripts in IceSL for 3D printing. This document guides the users to develop important components required to put together the Cycloidal Gear assembly with customized parameters.

This user manual is structured with an objective for the user to be able to flexibly interpret various parameters, analyze the performance, and also able to reproduce the entire demonstrated parametrical model in the Lua platform. Here also included in the documentation are the instructions for extracting necessary '.stl' files and G-codes allowing for cross-platforms implementations and 3D printing of the components.

The demonstrated model includes modelling of three main components of the geometry namely:

1. Base plate
2. Cycloidal gear
3. Eccentric cam



[Fig 1]: Cycloidal Gear Geometry Components

Script Overview

The multi-functional script allows the users to generate the entire assembly together with their inter-dependencies as well as individual components. The .lua file can be read/edited using text editors supported by IceSL Slicer or with IceSL-forge.

The scripting structure can be separated into following sections:

1. Parameter Definition
2. Gear Functions
3. Base Plate Functions
4. Eccentric Cam Functions

Parameter Definition

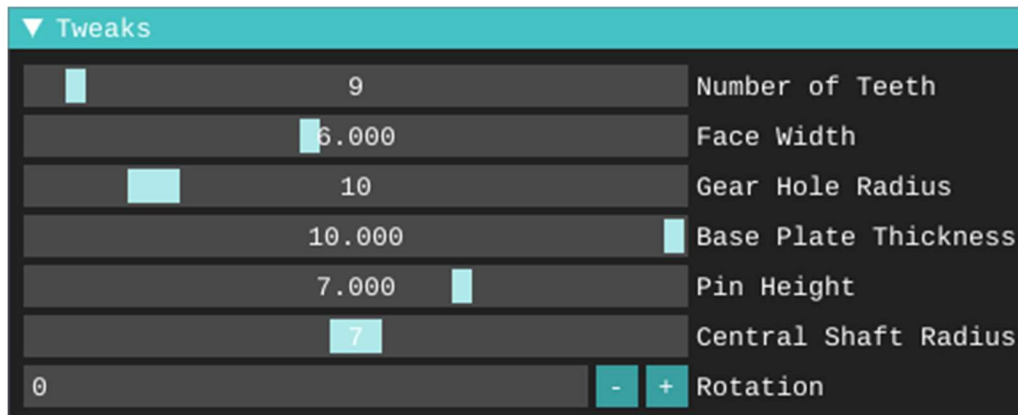
We are dividing the Parameter section into two parts:

- a) Input parameters: Defined by user
- b) Inter dependent parameters: Derived from formulas and input parameters.

Input parameters

The first and most important stage in modeling is defining the fundamental parameters. Users can provide the input values to the Tweak Box [Fig. 2] for developing custom models for simulation which provides real-time modifications and analysis.

You may change the default setting in the Tweak Box by modifying the below section [Fig 2] in the Lua script.



```
-- Defining the parameters with user inputs
local Z = ui_number("Number of Teeth", 9, 7, 40) -- Number of Teeth
local N = Z + 1 -- Number of Pins (Rollers)
local R = Z * 3 -- Rotor Radius, scales with Z
local da = 1.8 -- Adjusted Shaft Eccentricity to ensure contact
local extrusion_depth = ui_scalar("Face Width", 6, 3, 10) -- Extrusion depth of the gear
local gear_hole_radius = ui_number("Gear Hole Radius", 10, 8, 20) -- Radius of the hole in the gear
```

[Fig 2]: Tweak Box Value Modification

Inter-Dependent Parameters

The interdependent parameters that are important for gear profile modeling are derived with the help of the input parameters. The derived parameters are utilized in the generation of gear-pair geometry as represented in the below figure.

```
-- Fixed parameters for the model
local dp = 11.5 -- Pin diameter
local base_plate_thickness = ui_scalar("Base Plate Thickness", 10, 1, 10) -- Thickness of the base plate
local pin_height = ui_scalar("Pin Height", 7, 1, 10) -- Height of the pins
local pin_radius = dp / 2 -- Radius of the pins
local central_shaft_radius = ui_number("Central Shaft Radius", 7, 1, 13) -- Radius of the central shaft
local central_shaft_height = base_plate_thickness + 2 -- Height of the central shaft

-- Derived parameters for calculations
local housing_circle_circumference = 2 * math.pi * R -- Circumference of the housing circle
local roller_radius = housing_circle_circumference / (4 * N) -- Roller radius
```

[Fig 3]: Fixed parameter of model

Gear Functions

This section of the code is responsible for calculating the coordinates that form the shape of the cycloidal gear. Imagine drawing a curve by plotting 720 points around a circle. Each point's position is determined by an angle, theta, which ranges from $-\pi$ (negative pi) to π (pi). To get these positions, the code uses the `cycloidal_path` function. This function takes an angle, theta, and calculates where the point should be on the X and Y axes. The calculations consider the gear's rotor radius, the radius of the rollers (pins), and how much the shaft is offset from the centre (eccentricity).

- i. Function `cycloidal_path`: This function is involved in creating the cycloidal curve.

```
-- Function to calculate the cycloidal path
function cycloidal_path(theta)
    local x = (R * math.cos(theta))
    local y = (-R * math.sin(theta))
    return x, y
end
```

[Fig 4]: Function for Cycloidal Path

- ii. This section generates the cycloidal gear from the above function.

```
-- Generate the cycloidal gear path
local theta = {}
local x = {}
local y = {}
for i = 0, 719 do
    theta[i + 1] = -math.pi + (2 * math.pi / 719) * i
    local angle = theta[i + 1]
    local X_val, Y_val = cycloidal_path(angle)
    table.insert(x, X_val)
    table.insert(y, Y_val)
end
```

[Fig 5]: Cycloidal gear path generation

- iii. Function create3DCycloidalGear:

This part defines a function that takes the 2D coordinates of the cycloidal path and extrudes them into a 3D shape to form the cycloidal gear. The function "create3DCycloidalGear" is defined and it starts by creating an empty list called points. The function then goes through all the X and Y coordinates that define the cycloidal path. For each coordinate, it adds a point to the points list. Once all the points are collected, the function uses linear_extrude to stretch this 2D outline upwards into a 3D shape. The height of this shape is determined by the extrusion_depth parameter.

```
-- Function to create a 3D cycloidal gear by extruding the 2D path
local function create3DCycloidalGear(X, Y, depth)
    local points = {}
    for i = 1, #X do
        table.insert(points, v(X[i], Y[i]))
    end
    return linear_extrude(v(0, 0, depth), points)
end
local cycloidalGear = create3DCycloidalGear(X, Y, extrusion_depth) -- Create the cycloidal gear
```

[Fig 6]: Function to create 3D cycloidal gear by extruding the 2D

Base Plate Functions

Base plate is the important supporting part for our Cycloidal gear assembly where with its support we demonstrate the rotation of our cycloidal gear through its roller pins.

- i. Function createBasePlate: This function is involved in modelling the base plate for the cycloidal gear assembly.

```
-- Function to create the base plate
local function createBasePlate(radius, thickness)
    return translate(0, 0, -thickness / 2) * cylinder(radius, thickness)
end
```

[Fig 7]: Function to create the base plate

ii. Function createPins:

This function creates the cylindrical pins that will be positioned on the base plate. These pins are crucial because they interact with the cycloidal gear, providing support and guiding its movement. The function "createpins" is defined and it starts by making an empty list called pins to keep track of all the pins we're going to create. The function goes through each (x, y) coordinate where a pin needs to be placed. These positions have been calculated earlier to ensure the pins are correctly spaced. For each coordinate, it uses the translate function to move to the correct position and the cylinder function to create a pin of the specified radius and height. Each pin is then added to the 'pins' list. Once all the pins are created, the function combines them into a single object using the union function. This combined object is then returned by the function.

```
-- Function to create pins at calculated positions
local function createPins(pin_positions, pin_radius, pin_height)
    local pins = {}
    for _, pos in ipairs(pin_positions) do
        local x, y = unpack(pos)
        table.insert(pins, translate(x, y, 0) * cylinder(pin_radius, pin_height))
    end
    return union(pins)
end
```

[Fig 8]: Function to create pins at calculate positions

Eccentric Pin Functions

- Function createEccentricCam: We will be using an eccentric cam and Input shaft that aids the rotation of the gear through the base plate pins.

```
-- Function to create the eccentric cam and input shaft
local function createEccentricCam(eccentric_radius, eccentric_height, input_height, offset)
    -- Create the larger part of the eccentric cam
    local eccentric_cam = translate(0, 0, -eccentric_height / 2) * cylinder(eccentric_radius, eccentric_height)

    -- Create the smaller input shaft, offset within the larger eccentric cam
    local input_shaft = translate(offset, 0, -eccentric_height / 2) * cylinder(input_radius, eccentric_height + input_height)

    -- Combine the larger and smaller parts
    return union({
        eccentric_cam,
        input_shaft
    })
end
```

[Fig 9]: Function to create eccentric cam and input shaft

Rotation and Translation

Following section is involved in the rotation and translation of our complete Gear assembly.

```
-- Rotation and Translation
local eccentric_cam_rotation = ui_numberBox("Rotation", 0) -- Eccentric cam rotation input
local gear_translation_x = da * math.cos(math.rad(eccentric_cam_rotation * 10)) -- Calculate gear translation in x
local gear_translation_y = da * math.sin(math.rad(eccentric_cam_rotation * 10)) -- Calculate gear translation in y
local gear_rotation = -eccentric_cam_rotation * 10 / Z
```

[Fig 10]: Function for rotation and translation

Emit the components

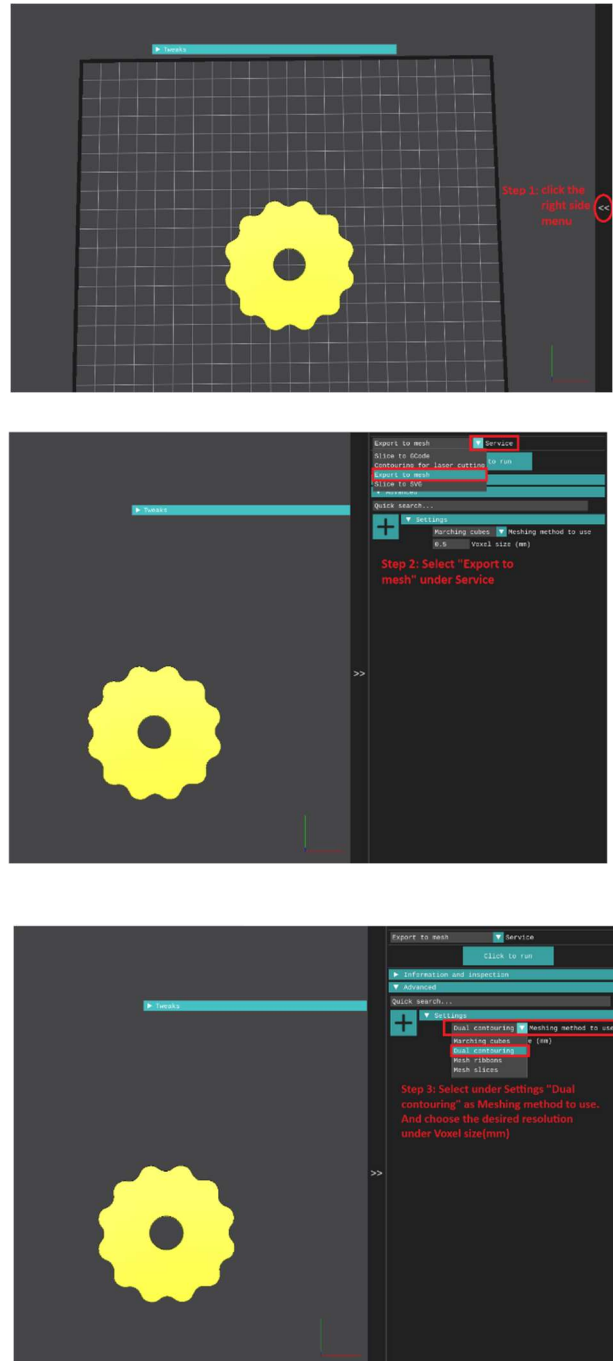
The following section emits all the components of the assembly together. In order to emit individual components we just need to execute the emit function of that particular component and all other part of the script remains the same.

```
-- Emit parts separately for 3D printing
emit(base_plate, 1) -- Emit base plate
emit(translate(0, 0, base_plate_thickness / 2) * pins, 2) -- Emit pins positioned correctly
emit(eccentric_cam, 3) -- Emit eccentric cam positioned correctly
emit(translate(gear_translation_x, gear_translation_y, base_plate_thickness / 2) * rotate(0, 0, gear_rotation) * finalGear, 4)
-- Emit cycloidal gear with translation and rotation applied
```

[Fig 11]: Emit

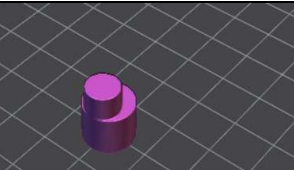
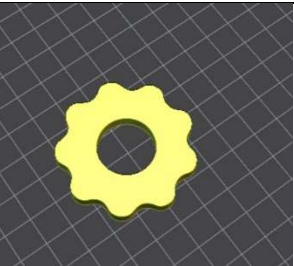
Extract the stl files

The following images shows us how to generate the .stl files for slicing in IceSL Slicer [Fig]



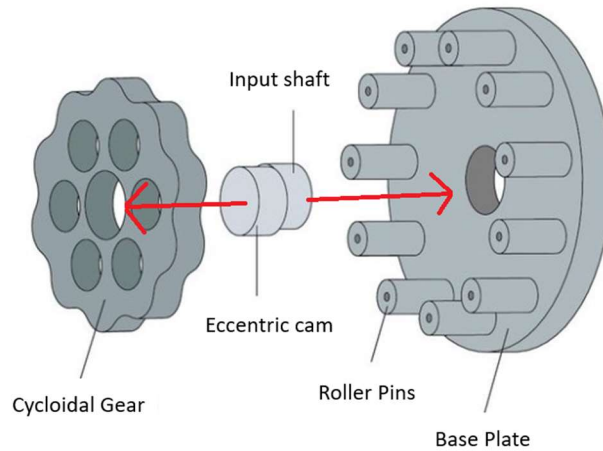
[Fig 12] : Steps to Extract stl files

Part List

No.	Name of the components	Process	Quantity	3D Image
1	Base plate	3D printing	1	 A 3D model of a circular base plate with a central hole and 12 smaller holes arranged in a circle. The model is cyan and is shown on a dark gray grid background.
2	Eccentric Pin	3D printing	1	 A 3D model of an eccentric pin, which is a cylindrical component with a central hole. The model is purple and is shown on a dark gray grid background.
3	Cycloidal Gear	3D printing	1	 A 3D model of a cycloidal gear, which is a gear with a central hole and 12 teeth. The model is yellow and is shown on a dark gray grid background.

Assembly

The Base plate with roller pins, Eccentric Pin and Cycloidal gear can be 3D printed as per the user defined parameters with the instructions stated above.



[Fig 13]: Assembly of the whole Geometry. Source: https://www.researchgate.net/figure/Exploded-view-of-the-different-parts-of-a-cycloid-drive-top-and-a-detailed_fig11_334030874

Steps:

- i. 3D print all the components separately
- ii. Fix the Input shaft to the base plate hole properly
- iii. Fix the Eccentric cam to the Cycloidal gear hole properly
- iv. The assembly is complete and check for proper alignment
- v. The rotation of the cycloidal gear is now possible